DAIFOOTCARE

On-Device Deep Learning for Diabetic-Foot-Ulcer Assessment

Model Design, Training, and Evaluation

A Technical Reference for the Three-Model Analysis Pipeline

Segmentation · Tissue Classification · Infection & Ischaemia

Prepared as supporting documentation for doctoral research

Version 1.0 · Last updated 2 July 2026

Contents

1 Overview and System Design

This document is the technical reference for the machine-learning components of **DaiFootCare**, a mobile application that performs automated assessment of diabetic foot ulcers (DFU) entirely **on-device**. Three deep-learning models cooperate in a single inference pipeline: a wound segmentation model that produces pixel-level masks and derived clinical measurements, a tissue-classification model that identifies the tissue types present in the wound bed, and an infection-and-ischaemia model that estimates two clinically-validated risk indicators. Each section below documents the purpose, data provenance, architecture, training procedure, evaluation protocol, results, and limitations of a model, at a level of detail intended to support reproduction and academic reporting.

1.1 Design principle: one shared backbone

The central architectural decision of the system is that **Models 2 and 3 share a single frozen CLIP ViT-B/32 image encoder** (`clip_backbone_fp16.tflite`). The backbone is the only large asset in the deployment. It is loaded once and executed once per input image to produce a 512-dimensional embedding; each downstream task then contributes only a small "head" network — 20 MB for tissue classification and 262 KB for infection/ischaemia. Model 1 (segmentation) is standalone. This factoring keeps the installed application small and keeps per-image inference fast, because the expensive encoder pass is amortised across two clinical tasks.

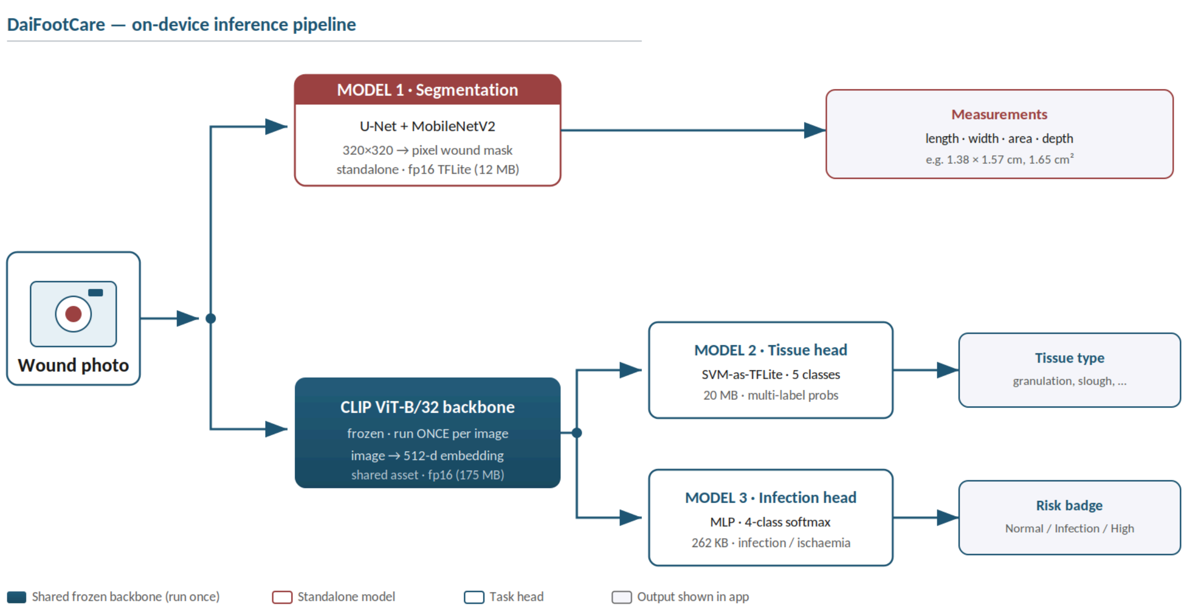

Figure 1. High-level dataflow. The CLIP backbone is evaluated a single time per image and feeds both classification heads; Model 1 runs independently to produce measurements.

1.2 Model summary

#	Model	Task	Backbone	Deployed file(s)	Headline result
1	Wound Segmentation & Measurement	pixel mask → length / width / area / depth	U-Net + MobileNetV2	model1_wound_fp16 .tflite (12 MB)	Test Dice 0.818 · val acc 0.89

#	Model	Task	Backbone	Deployed file(s)	Headline result
2	Tissue Classification	5 tissue types present in wound	CLIP ViT-B/32 (frozen) + SVM head	clip_backbone_fp16.tflite (175 MB) + tissue_head.tflite (20 MB)	AUC 0.825 / F1 0.733
3	Infection & Ischaemia	infection + blood-flow status	CLIP ViT-B/32 (frozen, shared) + MLP head	infection_ischaemia_head.tflite (262 KB)	Infection AUC 0.890 · Ischaemia AUC 0.987

Table 1. The three models, their tasks, backbones, deployed artefacts, and headline evaluation metrics.

1.3 Datasets used across the project

Five source collections are used across the three models: **DFUTissue** and **DFUTissueSegNet**[6] (finely-annotated tissue and wound masks), **DFUC2020** and **DFUC2021** (the Diabetic Foot Ulcer Challenge collections[7]), and **dfu-extra-labels**, a set of doctor-reviewed annotations added specifically for this project. The role each dataset plays differs by model and is documented in the relevant section. For the origin, licensing, and inter-relationship of the Challenge datasets, readers are referred to the official DFU dataset overview[7]; the specific Challenge papers required by the dataset providers are cited in full in Section 7.

2 Model 1 — Wound Segmentation & Measurement

Notebook: `model-1-kag.ipynb` (Kaggle, GPU). **Deployed artefact:** `model1_wound_fp16.tflite`.

2.1 Purpose and task

Model 1 locates the wound region in a photograph as a pixel-level mask and then converts that mask into clinical measurements: **length, width, area, and an estimated depth**. Formally the task is **binary semantic segmentation** — every pixel is classified as either wound or background. The network consumes an image tensor of shape `[1, 320, 320, 3]` (float32, values in `[0,1]`) and produces a per-pixel wound-probability mask of shape `[1, 320, 320, 1]`. A deterministic post-processing stage converts this mask into measurements.

2.2 Datasets

Dataset	Role	Notes
DFUTissue (Labeled / Original)	Primary target — precise pixel masks	≈75 finely-annotated images; the true supervision signal. Validation and test use only these, so reported metrics reflect genuine wound-boundary quality.
DFUC2020	Weak pseudo-masks (bounding boxes)	Coarse box-derived masks used for training only, capped at ≈250 examples to avoid pulling the model away from precise boundaries.
DFUC2021	Unlabeled images	Used solely to supply the representative dataset for int8 export; it is a classification set with no localisation labels.

Table 2. Data sources for Model 1 and the distinct role each plays in training and evaluation.

To make the most of the small precise set, the finely-annotated images (from **DFUTissue**[6], with the weak box masks drawn from **DFUC2020**[9,10]) are **oversampled ×3** relative to the box masks, and **strong augmentation** (horizontal/vertical flips, rotations, and random cutout/erasing) is applied to improve generalisation and counter overfitting on the roughly 75 precise examples.

2.3 Architecture and loss

- **Backbone:** a U-Net decoder[1] built on a pretrained **MobileNetV2**[2] encoder (ImageNet transfer learning). Encoder skip connections are taken at input /2, /4, /8, /16 with the bottleneck at /32; the backbone input is internally rescaled to `[-1, 1]`. A 320×320 input (÷32) preserves finer wound-boundary detail than a 256×256 input.
- **Loss:** `0.5 × BCE + Focal-Tversky`[3]. The Tversky term uses $\alpha = 0.3$, $\beta = 0.7$, which penalises missed wound pixels more heavily to raise recall and Dice; the focal component ($\gamma = 0.75$) concentrates learning on the small wound region.

Training schedule (multi-phase)

1. **Phase 1 —** freeze the encoder and train the decoder only (25 epochs, learning rate 1e-3).
2. **Phase 2 —** fine-tune the top 36 of 155 encoder layers with batch-normalisation frozen (learning rate 5e-5).
3. **Self-training —** add high-confidence pseudo-labels and continue training; the best-of-both checkpoint is retained by validation loss.

Callbacks and regime: `ModelCheckpoint` (best `val_loss`), `ReduceLROnPlateau`, and `EarlyStopping` with best-weight restoration. Mixed-precision training is used on GPU.

2.4 Measurement post-processing

The mask-to-measurement conversion is deterministic and is ported one-to-one from the training notebook into the app's Dart code, guaranteeing that on-device measurements match the reference implementation:

4. **Binarise** the probability mask (default threshold 0.5, with a half-peak fallback if no pixel clears the threshold).
5. Apply a 5×5 morphological **open then close** to remove specks and fill small holes.
6. Retain the **largest connected component** as the wound.
7. **Length and width** are taken from the component bounding box and scaled to centimetres using a pixels-per-cm factor from reference-object calibration. If uncalibrated, the wider frame side is assumed to be 12 cm and `isCalibrated=false` is flagged. **Area** = wound-pixel count × scale².
8. **Depth (heuristic)** = wound-pixel density inside the bounding box × 1.5 mm. This is an estimate, not a true 3-D measurement.

2.5 Results

Two families of metrics are reported. First, **per-epoch training metrics** logged by Keras at the selected checkpoint (best validation loss, epoch 2 of the fine-tuning phase): pixel accuracy, intersection-over-union (IoU / Jaccard), and the differentiable soft-Dice coefficient. Second, the **final segmentation quality** computed after the full post-processing pipeline (binarisation, morphology, largest-component selection) on the held-out precise masks, which is the figure that reflects true wound-boundary quality.

Split	Pixel accuracy	IoU (Jaccard)	Dice
Training (selected checkpoint)	0.945	0.597	0.746
Validation (per-epoch, Keras)	0.892	0.526	0.689
Validation (final, post-processed)	—	—	0.706
Test (held-out, 8-view TTA)	**—**	**—**	**0.818**

Table 3. Segmentation performance. Training/validation pixel accuracy, IoU and soft-Dice are per-epoch Keras metrics at the selected checkpoint (epoch 2). The final post-processed validation Dice (0.706) and the held-out test Dice (0.818, 8-view test-time augmentation) are the primary quality figures. Best mask threshold on validation: 0.58.

READING THE METRICS

Pixel accuracy is high (≈0.89–0.95) partly because background pixels dominate the frame, so a model is “correct” on most pixels before locating the wound at all. For this reason **Dice and IoU are the primary segmentation metrics** and pixel accuracy is reported only as a supporting figure. The per-epoch (soft) Dice and the final post-processed Dice differ slightly (0.689 vs. 0.706 on validation) because the latter is computed after binarisation and morphological clean-up rather than on the raw probability map.

Example output: `{ length 1.38 cm, width 1.57 cm, area 1.65 cm², depth 0.12 cm }`.

2.6 Exported artefacts

- `model1_wound_fp16.tflite` — **12.1 MB** — deployed in the application.

- `model1_wound_int8.tflite` — 6.5 MB — a smaller alternative; int8 quantisation makes the derived measurements noisier.

2.7 Limitations

LIMITATIONS OF MODEL 1

Depth is a pixel-density heuristic, not a genuine 3-D measurement. Absolute centimetre values require a calibration reference object in the frame; without one, only the relative **area trend over time** is reliable. The precise-mask training set is small (≈75 images), which augmentation and the weak DFUC2020 box masks only partly compensate for.

3 Model 2 — Tissue Classification

Notebook: `dfut-model3-newdata-gold.ipynb`. **Deployed:** `clip_backbone_fp16.tflite` + `tissue_head.tflite` (+ `tissue_head_meta.json`).

3.1 Purpose and task

Model 2 identifies which **tissue types** are visible in the wound bed. Because several tissue types can be present simultaneously, this is a **multi-label classification** problem rather than a single-label one. A CLIP-preprocessed wound image is mapped to a 512-dimensional embedding, from which five independent probabilities are produced. A tissue is judged **present** when its probability meets or exceeds its individually-tuned threshold.

Classes (5): `epithelial`, `granulation`, `necrosis`, `callus`, `slough`. The `fibrin` class was **dropped** because the clean evaluation set contained only two positive examples, too few to score reliably.

CLASSIFICATION, NOT SEGMENTATION

Model 2 answers “is each tissue type present in this wound?” at the image level. It does **not** produce pixel masks of tissue regions — that distinction matters when interpreting its outputs against the pixel-level Model 1.

3.2 Datasets

Dataset	Role
DFUTissue	Tissue-labeled wound images.
Source A doctor labels (dfu-extra-labels)	367 doctor-reviewed images (…orignalimagereNamed….xlsx).
DFUC2020 doctor labels (dfuc2020_annotations-4.csv)	810 training images carrying real 6-class doctor labels (previously only bounding boxes).

Table 4. Tissue-classification data sources. Doctor-reviewed labels were added specifically for this project.

Pooling these sources yields a **gold set of 1,176 doctor-labeled images**; training is further expanded by augmentation. Evaluation is reported on the 366 Source-A images so that results are directly comparable to the historic best baseline.

3.3 Architecture

- **Encoder:** a **frozen CLIP ViT-B/32**[4] image encoder producing a 512-d embedding. This embedding carries the predictive power — a from-scratch CNN scored near-random (≈0.55 AUC).
- **Head:** a per-class **RBF-SVM** trained on the CLIP embeddings.
- **Mobile export:** the SVM is reproduced exactly as TensorFlow operations (“SVM-as-TFLite”), verified to within 8e-6 of the desktop SVM, and exported as `tissue_head.tflite`.
- **Research vs. mobile:** the research-best desktop variant rank-averages **CLIP** ⊕ **DINOv2**[5], but DINOv2 (ViT-L, ≈1.2 GB) is impractical on phones, so the deployed mobile build is **CLIP-only**.

3.4 Results (Source-A evaluation)

Variant	AUC	F1
Historic best	0.835	0.656

Variant	AUC	F1
Desktop best (CLIP ⊕ DINOv2 ensemble)	0.838	0.726
Mobile (CLIP + SVM-as-TFLite, deployed)	**0.825**	**0.733**

Table 5. Tissue-classification results. The deployed mobile model exceeds the historic best on F1 by +0.077 while running fully on-device.

3.5 Output and thresholds

The model ships **raw probabilities**; the application applies per-class thresholds read from `tissue_head_meta.json`: epithelial 0.09, granulation 0.43, necrosis 0.60, callus 0.45, slough 0.63. The stored cross-validation metrics are AUC 0.825 and F1 0.7327.

3.6 Files

File	Size	Role
clip_backbone_fp16.tflite	175 MB	image (1,224,224,3) ⟶ 512-d embedding (shared with Model 3)
tissue_head.tflite	20 MB	512-d ⟶ 5 probabilities
tissue_head_meta.json	—	class order, thresholds, cross-validation metrics

Table 6. Deployed artefacts for Model 2.

3.7 Limitations

LIMITATIONS OF MODEL 2

The mobile build is CLIP-only (no DINOv2), trading a small accuracy margin for a large reduction in size. The `fibrin` class is not scoreable because too few labelled examples exist.

4 Model 3 — Infection & Ischaemia

Script: `train_model3_infection.py`. **Deployed:** shared CLIP backbone + `infection_ischaemia_head.tflite` (+ `infection_ischaemia_head_meta.json`). Full detail in `MODEL3_INFECTION_HANDOFF.md`.

4.1 Purpose

Model 3 replaces an originally-planned "Pus Level" output, for which no pus-severity labels exist. Instead it predicts two clinically-validated indicators: **Infection** and **Ischaemia** (blood flow). The recognition of these two conditions from DFU images, and the underlying dataset design, follow the framework established by Goyal *et al.*[12] and the DFUC 2021 Challenge[8].

4.2 Task formulation

The model is trained as a **4-class softmax** over `[none, infection, ischaemia, both]` — the DFUC 2021 label structure[7,8]. From the softmax the application derives **two independent binaries**:

- `P(infection) = P(infection) + P(both)` → threshold **0.41** → Present / Not Present.
- `P(ischaemia) = P(ischaemia) + P(both)` → threshold **0.61** → Impaired / Adequate.

This derived-binary formulation handles the rare ischaemia class far better than a direct 4-way argmax. The two binaries feed two UI rows and a top-level **risk badge**: none → *Normal*, infection → *Infection Detected*, ischaemia → *Impaired Blood Flow*, both → *High Risk*.

4.3 Datasets

Datasets are pooled with one representative image per wound:

Dataset	Wounds	none / infection / ischaemia / both	Notes
DFUC 2021 (train.csv)	5,955	2552 / 2555 / 227 / 621	Whole-image labels; 3,994 unlabeled rows dropped.
Part B (DFUNet)	1,447	595 / 642 / 24 / 186	Jointly labeled (same wound in both Infection & Ischaemia folders) → maps to 4-class.
Pooled	**7,402**	**3147 / 3197 / 251 / 807**	**Part B lifts the scarce ischaemia / both classes.**

Table 7. Pooled infection/ischaemia dataset. Class counts show how Part B reinforces the under-represented ischaemia and both classes.

Part B's pre-made augmentation copies (`_M` mirror, `_R` rotations) are collapsed to one crop per wound: because the backbone is frozen, augments add no signal and risk leakage. Cross-validation uses **StratifiedGroupKFold(5) grouped by wound**, eliminating augment- and patch-level leakage.

4.4 Architecture

- **Encoder:** the **same frozen CLIP ViT-B/32**[4] as Model 2 (`clip_backbone_fp16.tflite`), verified byte-identical (MD5) to the backbone the head was trained against.
- **Head:** `L2-normalize → Dense(256, ReLU) → Dropout(0.5) → Dense(4, softmax)`, trained with **label smoothing 0.05** and class-balanced weights.

- **Model selection:** the head was chosen through two grouped-CV bake-offs (`model3_infection_experiments*.py`). The MLP beat linear logistic regression; label smoothing then beat weight-decay, focal loss, and a 5-seed ensemble — at **zero extra size** (a single 262 KB head).
- **Export:** fp16 TFLite; Keras↔TFLite parity max-abs-diff of 1.2e-04.

4.5 Results (5-fold grouped CV, out-of-fold)

Metric	Infection	Ischaemia
AUC	**0.890**	**0.987**
F1 @ tuned threshold	0.829	0.872
Threshold	0.41	0.61

Table 8. Out-of-fold cross-validated performance for the two derived binaries.

Aggregate performance: macro-binary **AUC 0.939**, macro-**F1 0.850**; 4-class accuracy 0.776. Per-source AUCs are near-identical across DFUC 2021 and Part B, confirming that pooling generalises.

Progression: linear logistic regression 0.838 / 0.963 (macro-F1 0.788) → MLP 0.885 / 0.985 (0.842) → **MLP + label-smoothing 0.890 / 0.987 (0.850)**.

4.6 Files

File	Size	Role
infection_ischaemia_head.tflite	262 KB	512-d embedding → 4-class softmax
infection_ischaemia_head_meta.json	—	classes, derived-binary map, thresholds, badge rule

Table 9. Deployed artefacts for Model 3. Note the head is under 0.3 MB.

4.7 Limitations

LIMITATIONS OF MODEL 3

Infection reflects broad clinical criteria (redness, warmth, pain, discharge), not exudate alone; neuropathic patients may under-present, so the output is not a sole diagnostic basis. The ischaemia AUC of 0.987 is near ceiling, whereas infection (0.890) retains the remaining headroom — the next levers are wound-crop preprocessing via Model 1 or additional Part B crops, both at essentially zero extra deployment size.

5 Shared Preprocessing

5.1 CLIP preprocessing (Models 2 & 3)

This pipeline must match CLIP exactly, or classification accuracy collapses:

9. Decode the image to **RGB**.
10. Resize so the **shorter side = 224** (bicubic).
11. **Centre-crop** to 224 × 224.
12. Divide by 255 → [0,1].
13. Per-channel normalise: mean = [0.48145466, 0.45782750, 0.40821073], std = [0.26862954, 0.26130258, 0.27577711].
14. Feed NHWC float32 [1,224,224,3], channel order RGB.

5.2 Model 1 preprocessing

Model 1 uses its own pipeline: resize to 320 × 320, divide by 255 → [0,1]. The [-1,1] rescale required by the MobileNetV2 backbone is baked into the exported graph.

6 Application Integration

- **Assets:** assets/models/ holds all four TFLite files (model1_wound_fp16.tflite, clip_backbone_fp16.tflite, tissue_head.tflite, infection_ischaemia_head.tflite) plus the two *_meta.json files.
- **Service:** AiService (lib/features/wound/analysis/services/ai_service.dart) loads all models, runs Model 1 for measurements, and runs the CLIP backbone **once** for both heads.
- **Result model:** AnalysisResult carries the measurements, tissueType, infection, ischaemia, and riskBadge.
- **UI:** the results screen shows the risk badge plus tissue type, Infection, and Blood-Flow rows; history/persistence (SQLite, DB v4) stores infection/ischaemia; the interface is EN/AR localised.
- **Offline verification:** python mobile_export/reference_inference.py <image.jpg> runs the backbone once and prints both heads' outputs.

7 References

References are grouped into methods/architectures and datasets. The dataset entries [7]–[12] are the citations mandated by the dataset providers (Manchester Metropolitan University, Diabetic Foot Ulcer Challenge) and are reproduced as officially required; they must be cited in any publication that uses the DFUC collections. Adapt the formatting to the citation style required by your institution (the entries below follow a plain numbered convention).

Methods and architectures

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2015*, LNCS vol. 9351, Springer, 2015, pp. 234–241. arXiv:1505.04597.

[2] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510–4520. arXiv:1801.04381.

[3] N. Abraham and N. M. Khan, "A Novel Focal Tversky Loss Function with Improved Attention U-Net for Lesion Segmentation," in *Proc. IEEE 16th Int. Symp. Biomedical Imaging (ISBI)*, 2019, pp. 683–687. arXiv:1810.07842.

[4] A. Radford, J. W. Kim, C. Hallacy, *et al.*, "Learning Transferable Visual Models from Natural Language Supervision" (CLIP), in *Proc. 38th Int. Conf. Machine Learning (ICML)*, PMLR vol. 139, 2021, pp. 8748–8763. arXiv:2103.00020.

[5] M. Oquab, T. Darcet, T. Moutakanni, *et al.*, "DINOv2: Learning Robust Visual Features without Supervision," 2023. arXiv:2304.07193.

Datasets

[6] M. K. Dhar *et al.*, "Wound Tissue Segmentation in Diabetic Foot Ulcer Images Using Deep Learning: A Pilot Study," 2024. arXiv:2406.16012. (DFUTissue dataset; code and data: github.com/uwm-bigdata/DFUTissueSegNet.)

[7] M. H. Yap, C. Kendrick, N. D. Reeves, M. Goyal, J. M. Pappachan, and B. Cassidy, "Development of Diabetic Foot Ulcer Datasets: An Overview," in *Diabetic Foot Ulcers Grand Challenge*, 2021, pp. 1–18. arXiv:2201.00163. doi:10.1007/978-3-030-94907-5_1.

[8] M. H. Yap, B. Cassidy, J. M. Pappachan, C. O'Shea, D. Gillespie, and N. D. Reeves, "Analysis Towards Classification of Infection and Ischaemia of Diabetic Foot Ulcers," in *2021 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, IEEE, 2021, pp. 1–4.

[9] B. Cassidy, N. D. Reeves, J. M. Pappachan, D. Gillespie, C. O'Shea, S. Rajbhandari, A. G. Maiya, E. Frank, A. J. Boulton, D. G. Armstrong, and B. Najafi, "The DFUC 2020 Dataset: Analysis Towards Diabetic Foot Ulcer Detection," *touchREVIEWS in Endocrinology*, vol. 17, no. 1, p. 5, 2021.

[10] M. Goyal, N. D. Reeves, S. Rajbhandari, and M. H. Yap, "Robust Methods for Real-Time Diabetic Foot Ulcer Detection and Localization on Mobile Devices," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 4, pp. 1730–1741, 2018.

[11] M. Goyal, N. D. Reeves, A. K. Davison, S. Rajbhandari, J. Spragg, and M. H. Yap, "DFUNet: Convolutional Neural Networks for Diabetic Foot Ulcer Classification," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 4, no. 5, pp. 728–739, 2018.

[12] M. Goyal, N. D. Reeves, S. Rajbhandari, N. Ahmad, C. Wang, and M. H. Yap, “Recognition of Ischaemia and Infection in Diabetic Foot Ulcers: Dataset and Techniques,” *Computers in Biology and Medicine*, vol. 117, p. 103616, 2020.

NOTE ON CITATIONS

Entries [7]–[12] are the citations required by the DFUC dataset providers and should be reproduced in full in any publication using these datasets; the overview paper [7] documents the origin and interconnection of the Challenge collections. Still to be added by the author: a citation/approval statement for the doctor-annotated `dfu-extra-labels` set, and, if reproduced from a specific implementation, a source for the CLIP preprocessing constants. The datasets are released for non-commercial research use only — confirm each licence before publication.

End of technical reference — DaiFootCare model documentation v1.0